

SUNAM KUNDAL

AI/ML Intern at SiMa.ai

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WORK EXPERIENCE

SiMa.ai – AI/ML Intern

Bengaluru, Karnataka

Jan 2026 – Present

- Compiled and deployed **on-device LLM models** on embedded edge silicon using internal **C/C++ SDKs** under Linux, validating **production-ready inference execution** on SiMa.ai MLSoC hardware targets.
- Engineered **Bash automation frameworks** for end-to-end GenAI feature validation – covering pre-quantized LLMs (**LLM-Compressor**), **Dynamic LoRA**, and **Mixture-of-LoRA Experts (MoLE)** – reducing manual validation effort by **40%+**.
- Profiled **inference and compilation pipelines** using **pytest** to identify hardware-level bottlenecks, improving execution throughput on embedded processors; integrated into **Jenkins CI/CD** pipelines for automated firmware-level AI feature validation.

Xerzer Developments Private Limited – Software Development Intern (NLP)

Remote

Dec 2024 – Feb 2025

- Improved **NLP intent classification** accuracy by **18–25%** through systematic **feature engineering**, hyperparameter tuning, and model selection using **Scikit-learn** and **spaCy**.
- Designed and implemented an **end-to-end ML classification pipeline** across **27 e-commerce query categories**, achieving **91% accuracy** on 100+ held-out test queries; evaluated with **precision, recall, and F1-score**; deployed experiments on **Google Cloud**.

PROJECTS

Iris Biometric Template Security System | *PyTorch, OpenCV, Fuzzy Vault, Paillier HE*

- Designed a **3-layer biometric security pipeline** (ResNet18 + Fuzzy Vault + Paillier Homomorphic Encryption) on **20,000 NIR images** across 1,000 subjects, achieving **95.8% GAR at 1.0% FAR** on the CASIA-Iris-Thousand benchmark.
- Fine-tuned **ResNet18** with **Transfer Learning** and **CenterLoss** in PyTorch to extract **256-dimensional iris embeddings**; applied Fisher discriminant quantization achieving **100-bit cryptographic security** (attack space $\approx 10^{31}$).
- Implemented **enrollment and authentication pipeline** with **OpenCV** (Daugman normalization, CLAHE) and 1024-bit asymmetric key management, satisfying **ISO/IEC 24745** biometric template protection requirements.

Rent a Ride — Full-Stack Car Rental Marketplace | *React, Node.js, Express, MongoDB, Razorpay, JWT* [GitHub](#) | [Live](#)

- Built a 3-role (User/Admin/Vendor) **MERN platform** with **40+ RBAC-secured REST endpoints** supporting vehicle booking, fleet management, and admin workflows.
- Implemented **Razorpay HMAC-SHA256 signature verification** and idempotency-key-based **duplicate-payment guards**, preventing replay attacks and eliminating double-charge risk on every transaction.

CampusSphere — AI-Powered Academic Platform | *React.js, Convex, Firebase Auth, Gemini API*

[GitHub](#) | [Live](#)

- Developed and shipped a **production full-stack platform** with **100+ registered users** (tracked via DB records), enabling real-time peer collaboration and structured academic note sharing.
- Integrated **Gemini LLM API** with **Convex vector search** for **RAG-based PDF analysis**, enabling semantic retrieval across user-uploaded notes without external vector infrastructure.
- Enforced **role-based access control** via **Firebase Auth** with role-scoped data permissions; optimized **Convex real-time queries** to minimize client re-renders and reduce unnecessary API calls.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript, SQL, Bash

AI/ML & Deep Learning: PyTorch, Scikit-learn, HuggingFace Transformers, OpenCV, Transfer Learning, Fine-tuning, NLP, RAG, Generative AI

Edge AI & LLMs: On-Device LLM Deployment, LLM-Compressor, Quantization, ONNX, Dynamic LoRA, MoLE

Backend & Web: Node.js, Express.js, REST APIs, React.js, MongoDB, MySQL, Firebase, JWT, RBAC

Systems & DevOps: Linux (Ubuntu/Embedded), Jenkins (CI/CD), Docker, Git, pytest, Bash Scripting, Embedded Hardware Validation

Cloud & Tools: Google Cloud Platform, NumPy, Pandas, spaCy, VS Code, Vercel

EDUCATION

National Institute of Technology, Srinagar

Bachelor of Technology in Computer Science and Engineering (CGPA: 7.91 / 10.0)

Jammu & Kashmir

2022 – 2026

ACHIEVEMENTS

- Ranked 13th** out of 50+ teams – **IIT BHU Machine Learning Hackathon**.
- Solved **750+ DSA problems** across LeetCode, GFG, and CodeChef; **LeetCode Rating: 1676** (Top 20%).